Problem Set 1

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Problem 1.1 (Sampling distribution). Consider the population $U = \{1, 2, \ldots, 8\}$ of size $N = 8$. The values of the variable y are

```
y <- c(1, 2, 3, 3, 7, 8, 8, 8)
```

We are interested in the population total $t_y = \sum_{i \in U} y_i$. From a sample, estimate $\widehat{t}_y$ by $\widehat{t}_y = N\bar{y}$, where $\bar{y}$ is the sample mean.

a) Generate all simple random samples of size $n = 4$ from U without replacement. Denote the set of all such samples by $\mathscr{S}_4$. For each $s \in \mathscr{S}_4$, compute $\widehat{t}_y(s)$ and use the resulting values to obtain the sampling distribution of $\widehat{t}_y$.

   Hint. Use the functions `combn()` and `choose()`.

b) Using the results from part a), compute numerically

$$\mathbb{E}_p(\widehat{t}_y) = \sum_{s \in \mathscr{S}_4} p(s)\, \widehat{t}_y(s)$$

   and

$$\mathsf{var}_p(\widehat{t}_y) = \sum_{s \in \mathscr{S}_4} p(s) \left\{ \widehat{t}_y(s) - \mathbb{E}_p(\widehat{t}_y) \right\}^2,$$

   where $p(s) = 1/\binom{N}{n}$.

   Compare the numerical variance with the theoretical variance under simple random sampling without replacement:

$$\mathsf{var}_p(\widehat{t}_y) = N^2 \left(1 - \frac{n}{N}\right) \frac{\sigma_y^2}{n},$$

   where

$$\sigma_y^2 = \frac{1}{N-1} \sum_{i \in U} (y_i - \bar{y}_U)^2.$$

Problem 1.2 (Population Characteristics). We use the dataset `sepe.dat`, which contains data for a fictitious (sub-) population of firms:

```
> sepe <- read.table("https://stat.ethz.ch/Teaching/Datasets/cas-das/sepe.dat",
                     header = TRUE)
```

The dataset contains the number of full-time employees (`empl`), environmental protection expenditures (`expend`), and an indicator for industry class 31 (`code = 1`).

a) Compute the total environmental protection expenditures, the mean expenditures per firm, the expenditures per employee in the population, and the mean of the firm-level expenditures per employee.

b) Compute the population variance of the number of employees ${\sigma^2}_{\text{empl}}$, the population variance of expenditures ${\sigma^2}_{\text{expend}}$, and their correlation.

c) Compute the proportion P_{31} of firms in industry class 31.

Problem 1.3 (Simulation). We continue to use the dataset `sepe.dat` (see Problem 1.2). We repeatedly draw simple random samples without replacement from the $N = 800$ firms.

a) For samples of size $n = 200$, use Monte Carlo simulation to estimate the variance of the estimator of mean environmental protection expenditures per firm (variable: `expend`). Draw 1,000 samples, compute the sample mean for each sample, and calculate the variance of the 1,000 estimates. Also create a histogram of the estimates.

   Compare the Monte Carlo variance with the theoretical variance from Problem 1.2b) and report the coefficient of variation of the estimator.

   > **Hint.** Use `replicate(n_sim, expr)`, where `n_sim = 1000` and `expr` draws one sample and computes its sample mean. To reproduce the solution, use `set.seed(11)` before drawing the sample.

b) Repeat the simulation for samples of size $n = 50$. Compare the coefficient of variation with the result from part a). Also create a histogram of the estimates.